

Three Functions in L^AT_EX

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1 Gradient Descent

Machine learning runs on this one line. The parameters θ move against the gradient of the loss L , scaled by the learning rate η .

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \quad (1)$$

2 Softmax

Softmax turns a vector of raw scores z into probabilities. Each entry is exponentiated, then divided by the sum of all exponentials, so the outputs are positive and sum to one.

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (2)$$

3 Position as a Function of Time

The position of a body under constant acceleration is a second order Taylor expansion around the starting time t_0 . The first derivative is velocity and the second derivative is acceleration.

$$x(t) = x(t_0) + \left. \frac{dx}{dt} \right|_{t_0} (t - t_0) + \frac{1}{2} \left. \frac{d^2x}{dt^2} \right|_{t_0} (t - t_0)^2 \quad (3)$$

With the usual shorthand x_0 for initial position, v_0 for initial velocity, a for acceleration, and the clock started at $t_0 = 0$, it becomes the version from every physics course.

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2 \quad (4)$$

4 Why These Three

Equation 1 practices subscripts and Greek letters, Equation 2 practices fractions with big operators, and Equations 3 and 4 practice derivatives and evaluation bars. Rebuild all four from scratch without looking, and math mode is yours.